Ensembles

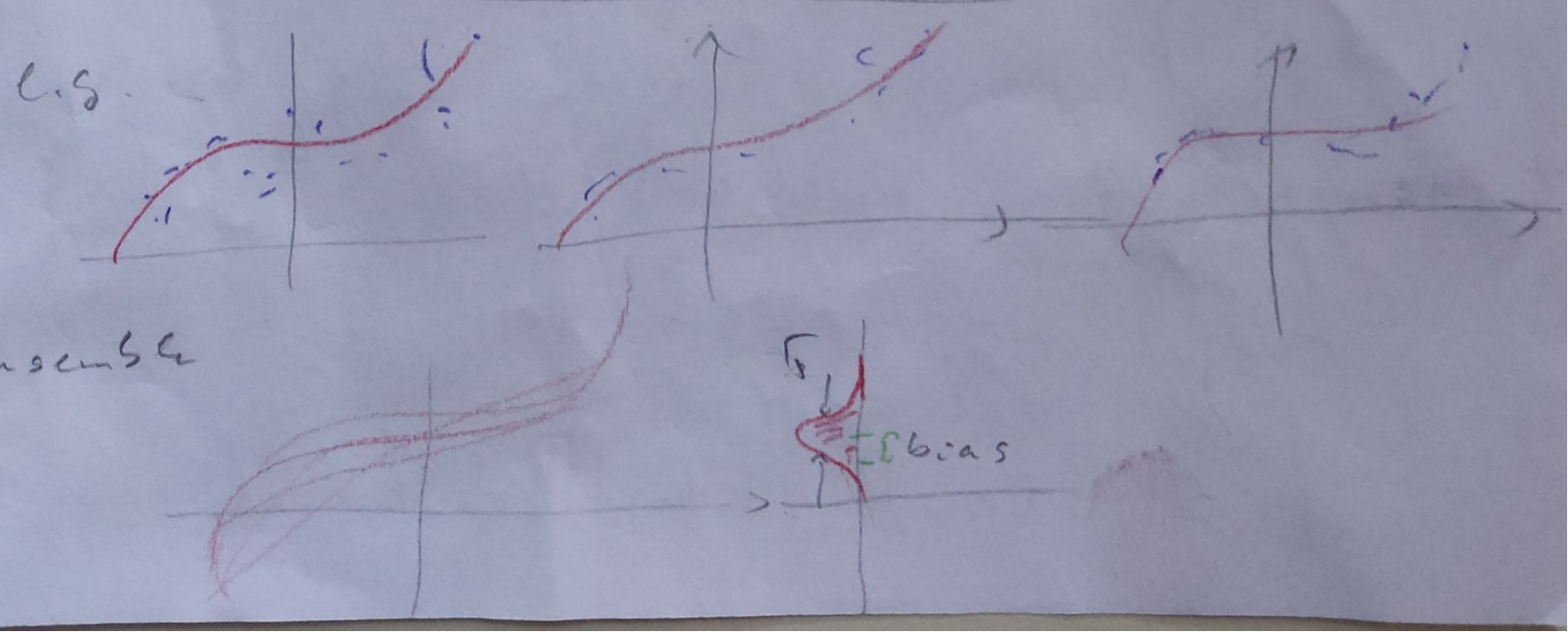
Combine models together work better
(- Bagging and boosting)

What can be an ensemble?
- Different algorithm
- Different choice of hyperparameters
- Different training data
- Training data with different weight

Bagging: train using independent random subsets of data

Boosting: Sequential trainings of classifiers focusing on previously misclassified data

Bias and variance of loss



Bias - variance decomposition

Loss function $\mathcal{L}(y, t) = \frac{1}{2}(y - t)^2$

Expected loss: $h(\underline{x})$ hypothesis, result of model

$$\mathbb{E}\left[(h(\underline{x}) - t)^2\right] = \mathbb{E}\left[(h(\underline{x}) - \mathbb{E}(h(\underline{x})) + \mathbb{E}(h(\underline{x})) - t)^2\right] =$$

$$= \mathbb{E}\left[(h(\underline{x}) - \mathbb{E}[h(\underline{x})])^2\right] + \left(\mathbb{E}[h(\underline{x})] - t\right)^2 +$$

$$+ 2\mathbb{E}\left[(h(\underline{x}) - \mathbb{E}[h(\underline{x})])(\mathbb{E}[h(\underline{x})] - t)\right] =$$

$$= \underbrace{\mathbb{E}\left[(h(\underline{x}) - \mathbb{E}[h(\underline{x})])^2\right]}_{\text{variance}} + \underbrace{\mathbb{E}\left[(\mathbb{E}[h(\underline{x})] - t)^2\right]}_{\text{bias}}$$

low variance — high variance

Illustration

low bias





high bias





Let us have m independent training data $h = \frac{1}{m}\sum_i h_i$

- Average / bias: unchanged

$$\mathbb{E}[h(x)] = \frac{1}{m}\sum_i \mathbb{E}[h_i(x)]$$

- Variance reduced

$$\mathrm{Var}[h(x)] = \mathrm{Var}\left[\frac{1}{m}\sum_i h_i\right] = \frac{1}{m^2}\sum \mathrm{Var}[h_i] = \frac{1}{m}\mathrm{Var}[h_i]$$

No independent datasets ...

Bootstrap aggregation or bagging

- Dataset D with n examples
- m new datasets D_i by replacement sampling of D
- Average the models

Problem: they are not independent

δ is the average correlation between datasets

$$\mathrm{Var}\left[\frac{1}{m}\sum_i h_i(x)\right] = \frac{1}{m}(1-\delta)\sigma^2 + \delta\sigma^2$$

Random forest: bagged decision trees

- bagged data
- at each branch only a random subset of features

Parameters importance:

- out-of-bag data
  - * features are permuted
  - * larger variance in out-of-bag data means more important data

- Probably the best black-box ML method

Boosting

- Weighted training set

Misclassification error $Err = \frac{1}{n} \sum_i I[h(x^{(i)}) \neq t^{(i)}]$

Use different weights for different data

$$Err^{(w)} = \sum_i w^{(i)} I[h(x^{(i)}) \neq t^{(i)}] \qquad w^{(i)} > 0, \quad \sum_i w^{(i)} = 1$$

Weak learner: e.g. Decision tree with low depth (even $n=1$!)

predicts data with prob. $1/2 + \varepsilon$ (ε can be small)

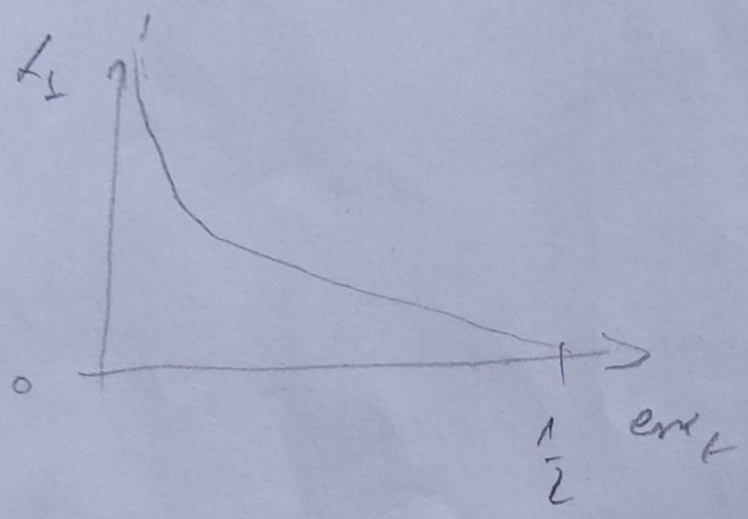
Adaboost

1. At each iteration (a) re-weight training data by assigning larger weight to misclassified data
2. Train classifier again
3. Add classifier to ensemble
4. Go to 1.

Adaboost: $y^{(i)} = \pm 1$

Error: $err_t = \sum_i w^{(i)} \mathbb{I}\left[h_t(x^{(i)}) \neq y^{(i)}\right]$

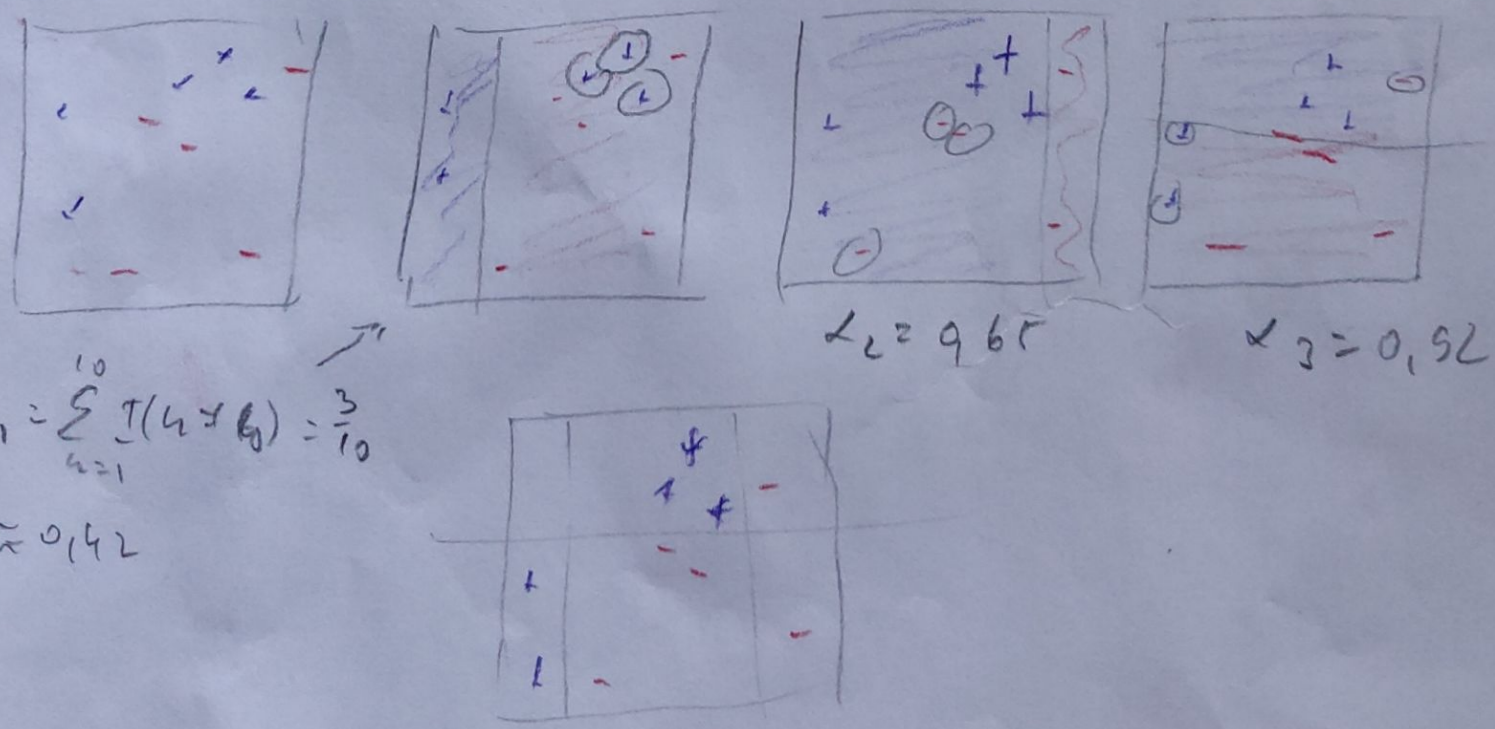
Classifier coefficient: $\alpha_t = \frac{1}{2} \log \frac{1 - err_t}{err_t}$



$w^{(i)} = w^{(i)} \exp\left(-\alpha_t y^{(i)} h_t(x^{(i)})\right)$

Misclassified: $h \neq y$ different sign $\Rightarrow$

$\Rightarrow \exp(+\ldots)$ w increases



$err_1 = \sum_{i=1}^{10} \mathbb{I}(h \neq y) = \frac{3}{10}$

$\alpha_1 \approx 0,42$

Ensembles

Bagging

- Reduces variance — robust to outliers
- Bias is not changed
- Parallel
- Reduces overfitting

Boosting

- Reduces bias
- Increases variance
- May overfit
- Sequential